



DOC ID: SL-POP-ERP-MS-001

BA Reference: DOC-001 v1.0 • Enterprise Edition

POPULAR AUTO SPARE & A/C PARTS CO. W.L.L

MODULE 01 OF 09

CORE CATALOGUE & ITEM ENGINE

Module 1: PCode Generation, Cataloguing & Item Master Engine

Comprehensive enterprise technical scope, dedicated engineering team matrix, 10-week deliverable breakdown, and milestone payment agreement for Popular Auto Spare & A/C Parts Co. W.L.L.

TOTAL DURATION

10 Weeks

CALENDAR SPAN

2.50 Months

TOTAL INVESTMENT

BD 3,809.091

PAYMENT MODEL

4 Gates (25%)

01

Section 1.0: Executive Summary & Multi-Module Roadmap Overview

MASTER CONTEXT

Module 01: PCode Generation, Cataloguing & Item Master Engine forms the master product cataloguing backbone of the Popular ERP transformation. Designed directly from the **DOC-001 v1.0 Business Analysis Document**, it eliminates unverified Excel-based item creation, enforces sequential FIFO PCode generation, automates multi-stage CTO technical approvals, and provides direct cross-reference compatibility intelligence for over 50,000 active SKUs across Bahrain and Dubai wholesale branches.

 **Complete 9-Module ERP Implementation Architecture (Module 01 Highlighted in Blue)**

 **MODULE 01 • CURRENT FOCUS**

PCode & Item Master Engine

FIFO sequential PCode numbering, 4-stage CTO approval, temporary item hold, cross-reference mapping, PartSouq API & multi-tier pricing.

10 Wks • 4 Gates

BD 3,809.091

MODULE 02

Vendor & Purchase Lifecycle

Global supplier cataloguing, 3-way RFQ procurement, landed cost tracking, custom clearance & automated replenishment.

12 Wks • 4 Gates

BD 4,570.909

MODULE 03

Store Verification & Stock Control

Inward store verification (PVN), blind 60/40 cycle counts, 5-tier bin numbering, 7-stage GIT transfers & quarantine management.

15 Wks • 4 Gates

BD 5,713.636

MODULE 04

Multi-Branch Sales & POS Engine

Multi-channel POS, customer credit limits, multi-tier discount approval, quotation pipeline & return management.

15 Wks • 4 Gates

BD 5,713.636

MODULE 05

Financial Accounting & VAT

Multi-currency general ledger, AP/AR automation, National Bureau for Revenue (NBR) VAT returns & audit trials.

16 Wks • 4 Gates

BD 6,094.545

MODULE 06

Enterprise Administration & Roles

Centralized multi-company configuration, RBAC, division isolation, system audit vault & automated backups.

14 Wks • 4 Gates

BD 5,332.727

MODULE 07

Human Resource Management

Bahrain LMRA/SIO compliance, automated WPS payroll processing, leave accrual, document expiry & EOSB calculations.

10 Wks • 4 Gates

BD 3,809.091

MODULE 08

Hardware & Infrastructure Setup

Local branch failover server architecture, thermal barcode printer mesh, high-speed PDA Wi-Fi roaming & UPS power backup.

08 Wks • 4 Gates

BD 3,047.273

MODULE 09

Executive BI & Analytics Dashboard

Real-time GM executive command center, branch turnover analysis, aging inventory heatmap & AI-driven predictive replenishment.

08 Wks • 4 Gates

BD 3,047.273

02

Section 2.0: Comprehensive Scope Matrix & Technical Architecture

DOC-001 REQUIREMENTS

Synthesized directly from the **DOC-001 v1.0 Business Analysis Document**, Module 1 addresses the core cataloguing challenges of duplicate entries, unverified OEM mapping, manual pricing updates, and third-party Excel dependencies across all 6 branches.



FIFO Sequential PCode Engine

Automated First-In-First-Out sequential code allocation (e.g. P000001 to P999999). Strictly eliminates manual code typing and eliminates duplicates with zero collision concurrency locks.



4-Stage CTO Approval Workflow

Formal request queue where branch managers submit item requests, followed by CTO technical validation, OEM verification, pricing review, and global ERP inventory activation.



Exception Temporary Item Engine

Enables urgent intake of unlisted items under temporary code holding. Automatically locks transaction finalization until formal CTO approval, followed by automatic retrospective ledger synchronization.



Substitute & Cross-Reference Matrix

Intelligent OEM compatibility engine mapping alternate part numbers, aftermarket equivalents, and direct replacements, giving sales counters instant substitute recommendations during stockouts.



PartSouq External Catalog Integration

Deep link integration with external automotive parts reference repositories (PartSouq) allowing technical sales teams to verify vehicle schematics, exploded diagrams, and OEM part fits in real-time.



Multi-Tier Pricing & MOL Control

Centralized pricing engine supporting Retail, Wholesale, Garage/Workshop, and Export price lists with floor-margin protection, paired with automated Minimum Order Level (MOL) stock alert thresholds.

03

Section 3.0: Dedicated Engineering Team Cost & Resource Allocation Matrix

TRANSPARENT RATES

Pricing for Module 1 is determined directly by allocating a 5-person dedicated engineering team over **10 working weeks (2.50 calendar months)**. Rates are computed on standard 160-hour monthly engineering allocations in Bahraini Dinars (BHD):

Project Manager & Solutions Architect

BD 545.455

BD 136.364 / Wk • 2.50 Mo = BD 1,363.638

100% Dedicated

Back-End Lead Engineer (PHP MVC / APIs)

BD 227.273

BD 56.818 / Wk • 2.50 Mo = BD 568.183

100% Dedicated

Front-End Lead Engineer (React 18 / UI)

BD 227.273

BD 56.818 / Wk • 2.50 Mo = BD 568.183

100% Dedicated

Database & Cloud DevOps Architect

BD 204.545

BD 51.136 / Wk • 2.50 Mo = BD 511.363

100% Dedicated

QA & System Test Automation Lead

BD 159.091

BD 39.773 / Wk • 2.50 Mo = BD 397.728

100% Dedicated

ENGINEERING DESIGNATION	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Enterprise Solutions Architect	BD 545.455	BD 136.364	BD 3.409	2.50 Months (10 Wks)	BD 1,363.638
Back-End Lead Engineer (PHP MVC / REST APIs / Engine)	BD 227.273	BD 56.818	BD 1.420	2.50 Months (10 Wks)	BD 568.183
TOTAL MODULE 1 DEDICATED ENGINEERING INVESTMENT				10 Working Weeks	BD 3,809.091

ENGINEERING DESIGNATION	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H)	EFFORT ALLOCATION	TOTAL COST (BHD)
Front-End Lead Engineer (React 18 / Item Master UI)	BD 227.273	BD 56.818	BD 1.420	2.50 Months (10 Wks)	BD 568.183
Database & Cloud Infrastructure Architect	BD 204.545	BD 51.136	BD 1.278	2.50 Months (10 Wks)	BD 511.363
QA & System Test Automation Lead Engineer	BD 159.091	BD 39.773	BD 0.994	2.50 Months (10 Wks)	BD 397.728
TOTAL MODULE 1 DEDICATED ENGINEERING INVESTMENT				10 Working Weeks	BD 3,809.091

04

Section 4.0: Detailed Milestone Breakdown & Deliverables (10-Week Roadmap)

4 DELIVERABLE GATES @ 25%

Module 1 delivery is structured across 4 sequential, independently verifiable engineering milestone tranches over a 10-week implementation lifecycle. Each milestone requires technical sign-off and UAT demonstration before gate payment authorization:

Milestone 1: Multi-Branch Hierarchy, PCode FIFO Engine & Group Taxonomy

🕒 Weeks 1-3 • 25% (BD 952.273)

Establishes the foundational enterprise organizational hierarchy (Bahrain HQ, 6 Automobile branches, 2 Industrial branches, 1 Service center, UAE wholesale) and builds the core sequential FIFO PCode generation engine with non-collision locks and multi-tier group taxonomy.

🔑 KEY ENGINEERING DELIVERABLES:

- ✓ Multi-Company & Multi-Branch entity data model with BHD/AED/USD currency and VAT settings.
- ✓ Sequential FIFO PCode auto-generation service (6-digit alphanumeric with collision prevention).
- ✓ Master Taxonomy Engine: Group, Sub-Group, Brand, Category, Make, Model, and Year trees.
- ✓ Centralized Item Master database schema with high-performance indexing for 50,000+ SKUs.
- ✓ Role-Based Access Control (RBAC) foundation for Cataloguing, Store, and Executive roles.

Acceptance Verification: 10,000 continuous sequential PCode generation stress test with 0 duplicate collisions; multi-branch entity isolation verified.

Gate Payment: BD 952.273

Milestone 2: 4-Stage CTO Approval Workflow & Exception Item Sync

🕒 Weeks 4–5 • 25% (BD 952.273)

Implements the complete governance workflow for item creation requests, empowering branch managers to request new products while enforcing CTO technical validation, temporary item holding locks, and retrospective ledger synchronization.

☰ KEY ENGINEERING DELIVERABLES:

- ✓ Branch Request Submission Portal with mandatory OEM, specifications, and photo attachments.
- ✓ CTO Technical Review Dashboard with duplicate detection, OEM validation, and approval tools.
- ✓ Temporary Item Code Generation Engine for urgent receipt with inventory dispatch hold lock.
- ✓ Retrospective Transaction Synchronization Service (replaces temp code across all pending documents upon CTO approval).
- ✓ Item Deletion Guardrails preventing deletion of any item linked to historic purchase/sales records.

Acceptance Verification: End-to-end simulation of 4-stage approval gate; temporary item hold lock verified to prevent unauthorized dispatch; retroactive sync tested.

Gate Payment: BD 952.273

Milestone 3: Cross-Reference Mapping, PartSouq API Integration & Smart Search

🕒 Weeks 6–8 • 25% (BD 952.273)

Integrates advanced automotive intelligence into the item master, including OEM inter-changeability mapping, PartSouq external parts diagram lookup, item handling sales warning alerts, and ultra-fast multi-parameter POS search.

☰ KEY ENGINEERING DELIVERABLES:

- ✓ OEM Inter-changeability & Direct Substitute Mapping Matrix per PCode.
- ✓ External Parts Catalog Integration: PartSouq deep linking and OEM diagram viewer integration.
- ✓ Item Handling Warnings & Sales Alerts (Fragile, Special Storage, HAZMAT, High-Value, Non-Returnable).
- ✓ Advanced Multi-Parameter Sales Search Engine (Fuzzy search by OEM, Brand, PCode, Arabic/English keywords).
- ✓ Substitute Recommender Widget embedded directly into sales quotation and invoice creation screens.

Acceptance Verification: Sub-second (<200ms) cross-reference search across 50,000 SKUs; PartSouq deep links active; sales warning alerts trigger modal locks.

Gate Payment: BD 952.273

Milestone 4: Multi-Tier Pricing, MOL Stock Alerts, Legacy PCode Archive & Final UAT

🕒 Weeks 9–10 • 25% (BD 952.273)

Deploys dynamic multi-tier pricing matrices, Minimum Order Level (MOL) automated reorder alerts, legacy PCode archiving and soft-hiding governance, multi-branch staging deployment, and comprehensive client UAT acceptance sign-off.

☰ KEY ENGINEERING DELIVERABLES:

- ✓ Multi-Tier Pricing Engine: Retail, Wholesale, Workshop/Garage, Export, and Minimum Floor Price.
- ✓ Centralized Price Update Engine with MD/CTO multi-level authorization workflows.
- ✓ Minimum Order Level (MOL) & Reorder Point (ROP) automated stock replenishment alert system.
- ✓ Legacy PCode Governance: Inactive PCode archiving and soft-hide from POS while preserving audit trails.
- ✓ Complete End-to-End System Testing, Multi-Branch Staging Deployment, User Training & Final UAT Sign-Off.

Acceptance Verification: All 4 pricing tiers functional across all 6 branches; legacy PCode soft-hidden; 100% pass rate in formal stakeholder UAT.

Gate Payment: BD 952.273

05

Section 5.0: Milestone Payment Structure & Invoicing Schedule (10-Week Model)

GUARANTEED MILESTONE GATES

Payments are strictly gated to verified milestone delivery and technical demonstration sign-off across 4 equal tranches of 25%:

INVOICE TRANCHE	TARGET TIMELINE	MILESTONE DELIVERABLE SCOPE	% VALUE	NET AMOUNT (BHD)	VAT 10% (BHD)	GROSS TOTAL (BHD)
Tranche 1 (Gate 1)	Week 3 Completion	Multi-Branch Entity Setup, PCode FIFO Sequence Engine & Group Taxonomy	25.00%	BD 952.273	BD 95.227	BD 1,047.500
Tranche 2 (Gate 2)	Week 5 Completion	4-Stage CTO Approval Workflow, Exception Item Code & Retrospective Sync	25.00%	BD 952.273	BD 95.227	BD 1,047.500
Tranche 3 (Gate 3)	Week 8 Completion	Cross-Reference Mapping, PartSouq API Integration & Smart Sales Search	25.00%	BD 952.273	BD 95.227	BD 1,047.500
Tranche 4 (Gate 4)	Week 10 Completion	Multi-Tier Pricing, MOL Stock Alerts, Legacy PCode Archive & Final UAT	25.00%	BD 952.273	BD 95.227	BD 1,047.500
TOTAL MODULE 1 INVESTMENT (10 WORKING WEEKS)			100.00%	BD 3,809.091	BD 380.909	BD 4,190.000

06

Section 6.0: Working Calendar, SLA, 15-Day Grace Period & Legal Terms

STANDARD SLA

1. 15-Day UAT Review SLA & Automatic Acceptance Grace Period

- Upon delivery of each milestone build to the staging environment, Popular Auto Spare will have a **15-calendar-day UAT review window** to test functionality and raise defect tickets.
- If no blocking issues are logged within 15 days, the milestone is deemed formally accepted, authorizing the corresponding invoice tranche release.

2. Source Code, Intellectual Property & Architecture Transfer

- All custom schemas, PHP MVC engines, React front-end components, and integration scripts developed for Module 1 become the exclusive intellectual property of Popular Auto Spare & A/C Parts Co. W.L.L upon final settlement.
- Zero ongoing vendor lock-in; complete developer documentation and database schemas will be handed over.

3. Defect Classification & Warranty Period

- **Severity 1 (Critical):** System halt or cataloguing blocker • Fix within 4 business hours.
- **Severity 2 (High):** Workflow or approval degradation • Fix within 12 business hours.
- **Severity 3 (Standard):** Cosmetic or minor search enhancement • Fix in scheduled sprint.
- **Warranty:** 60 calendar days of comprehensive bug-fix warranty following production sign-off.

07

Section 7.0: Stakeholder Authorization & Digital Sign-Off Console

MULTI-PARTY RATIFICATION

By signing below, authorized representatives ratify the technical scope, 10-week deliverable timeline, dedicated engineering team cost matrix, and milestone payment tranches for Module 01:

Client Authorized Signatory

Popular Auto Spare & A/C Parts
Co. W.L.L



Signatory: Managing Director /
General Manager

Solutions Architect Lead

SaNDS Lab Middle East W.L.L



Signatory: Enterprise Solutions
Architect

**Project Management
Advisor**

UniGlobal Business Solutions



Signatory: Strategic Project
Management Consultant